

Algebra First-Year Exam, January 2024, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Suppose k is a field.

- (i) If $A = k[X, Y]/(X^2Y^3)$, find all minimal prime ideals of A .
- (ii) Show that $B = k[X, Y]/(X^2, Y^3)$ has a unique prime ideal, and that this prime ideal is maximal. What is it?

2. Which of the following polynomials are irreducible? Explain.

- (a) $X^3 + X + 1$ in $\mathbb{F}_2[X]$;
- (b) $X^3 + X + 1$ in $\mathbb{Q}[X]$;
- (c) $X^5 + 10X^2 + 25X + 5$ in $\mathbb{Q}[X]$;

3. Suppose R is a ring with identity and M is a nonzero left R -module. Let $\text{End}_R(M)$ be the ring of R -linear maps $M \rightarrow M$. Show that if 0 and M are the only R -submodules of M , then $\text{End}_R(M)$ is a division ring.

4. Let R be an integral domain.

- (i) Define what it means for an element of R to be irreducible, and to be prime.
- (ii) Show that a prime element is irreducible.

5. Suppose F is a field, V and W are F -vector spaces of finite dimension and $f : V \rightarrow W$ is a surjective linear map, and let $U = \text{Ker}(f)$. Show using the definitions that $\dim_F(V) = \dim_F(W) + \dim_F(U)$.

6. Suppose L/K is an extension of fields. Show that $\alpha \in L$ is algebraic over K if and only if there is a nonzero sub- K -vector space $V \subseteq L$ of finite dimension such that $\alpha V \subseteq V$ (for “if,” pick a nonzero $v \in V$ and consider $\alpha^n v$ for $n \in \mathbb{N}$).

7. Find representatives of each conjugacy class of $GL_4(\mathbb{F}_2)$ of order 3 or 5 (the order of a conjugacy class is the order of any element of it).